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ORIGINAL RESEARCH

STRUCTURAL

Long-Term Outcomes of Patients Requiring Pacemaker Implantation After Transcatheter Aortic Valve Replacement

The SwissTAVI Registry

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ABSTRACT

BACKGROUND The impact of pacemaker (PM) implantation on outcomes following transcatheter aortic valve replacement (TAVR) remains controversial, especially as TAVR indications expand to low-risk patients.

OBJECTIVES This study sought to evaluate the all-cause and cardiovascular mortality of patients undergoing PM implantation after TAVR.

METHODS In this prospective, observational, nationwide TAVR cohort study, the outcomes of patients undergoing permanent PM implantation were investigated. Patients were enrolled from 19 centers across Switzerland between February 2011 and June 2022.

RESULTS Among 13,360 patients enrolled (mean age 82 ± 7 years, 47% female, self-expanding valves 48%, median follow-up 889 days [Q1-Q3: 365-1,765 days]), 2,028 (15%) required PM implantation within 30 days post-TAVR. Patients requiring post-TAVR PM implantation were older (82 ± 6 years of age vs 81 ± 7 years of age), were predominantly male (58% vs 50%), and more often had atrial fibrillation (34% vs 29%). At 1-year follow-up, these patients had higher overall mortality (aHR: 1.15; 95% CI: 1.05-1.26; $P = 0.002$) and cardiovascular mortality (aHR: 1.25; 95% CI: 1.06-1.46; $P = 0.006$). These trends persisted at 5- and 10-year follow-up. After multivariable adjustments, significantly higher rates of cardiovascular mortality, LVEF decline ≥10%, and NYHA functional class III or IV at 1-year follow-up were observed (aHR: 1.44,; 95% CI: 1.35-1.54; $P < 0.001$), along with higher all-cause and cardiovascular mortality rates at 5- and 10-year follow-up in patients requiring PM implantation following TAVR compared with those not needing a PM.

CONCLUSIONS In this large nationwide registry, patients receiving PM implantation within 30 days after TAVR had significantly higher rates of overall and cardiovascular mortality up to 10 years. (SwissTAVI Registry; NCT01368250) (JACC Cardiovasc Interv. 2025;18:1163–1171) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

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ABBREVIATIONS AND ACRONYMS

aHR = adjusted HR

HF = heart failure

LVEF = left ventricular ejection fraction

PM = pacemaker

TAVR = transcatheter aortic valve replacement

Transcatheter aortic valve replacement (TAVR) has become a first-line therapy for patients with severe symptomatic aortic stenosis, with its indication expanding to younger and low surgical risk patients.[1-3] Despite major advances in TAVR technologies and techniques, the most common post-TAVR complication remains conduction disturbances, especially high-degree atrioventricular block, potentially leading to syncope and/or cardiac death.[4-6]

Early identification of patients requiring pacemaker (PM) therapy after TAVR remains challenging. The 2021 European Society of Cardiology guidelines for cardiac pacing propose several risk stratification strategies, including as electrophysiological testing or prolonged ambulatory electrocardiography monitoring in patients with PR or QRS width prolongation.[7]

Appropriate selection of patients requiring PM implantation after TAVR is critical. It is essential to avoid missing patients at risk for life-threatening bradyarrhythmia while also preventing unnecessary early PM implantation after TAVR, which is associated with adverse right ventricular remodeling, reduced left ventricular ejection fraction (LVEF), and a higher incidence of heart failure (HF).[4-6]

Several individual studies, many of them retrospective, as well as a recent meta-analysis, have reported conflicting data of the impact of PM implantation after TAVR on clinical outcomes.[8-14] Despite the high incidence of atrioventricular conduction disturbances after TAVR, there is a paucity of data from large-scale studies assessing the long-term prognostic impact of PM therapy after TAVR.

As TAVR expands to include younger and low-risk patients with longer life expectancy, understanding the impact of PM implantation following TAVR becomes increasingly important. Therefore, we investigated the outcomes of patients requiring permanent PM implantation within the prospective, observational, nationwide, SwissTAVI (Swiss Transcatheter Aortic Valve Implantation) cohort study.

METHODS

STUDY DESIGN. The SwissTAVI registry is a prospective multicenter observational study gathering clinical, procedural, and follow-up data from successive patients undergoing TAVR implantation in Switzerland (NCT01368250). All Swiss heart valve centers (currently 19) are mandated by the Swiss Federal Office of Public Health to contribute to the SwissTAVI registry, enroll patients, and provide follow-up information. This registry is a collaborative effort led by the Swiss Working Group of Interventional Cardiology and the Swiss Society of Cardiac and Thoracic Vascular Surgery. The registry's rationale and design have been detailed in prior publications.[15-21]

Ethical approval was obtained from the national ethics committee and informed consent from all participants in accordance with the Declaration of Helsinki.

STUDY PARTICIPANTS. Patients undergoing TAVR implantation in 19 centers across Switzerland between February 2011 and June 2022 were included. For the purpose of the current study, only patients alive 30 days after TAVR have been included. Patients were divided into 3 groups: 1) patients with preexisting PM implantation prior to the TAVR procedure; 2) patients requiring PM implantation during the first 30 days after the TAVR procedure; and 3) patients without PM implantation.

DATA COLLECTION. A Web-based database employs standardized forms for data collection at baseline and during follow-ups, which are carried out according to a pre-established plan. Clinical events were recorded and evaluated by a dedicated clinical event committee on the basis of the second edition of the Valve Academic Research Consortium-2 standardized endpoint definitions. The Clinical Trials Unit at the University of Bern oversees central data monitoring to ensure data completeness, validity, and conducts independent statistical analysis.

OUTCOME DEFINITIONS. The primary objective of the study is to determine the impact of PM implantation on all-cause mortality at 1 year. Secondary

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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outcomes include cardiovascular death, a decrease in LVEF ≥10%, worsening in NYHA functional class III or IV at 1 year and up to 5 years of follow-up, and a composite outcome or the previously mentioned outcomes at 1- and at 5-year follow-up.

STATISTICAL METHODOLOGY. Continuous variables are presented as mean ± SD or median (Q1-Q3) and categorical variables as number and percentage. Groups were compared by the pairwise analysis of variance F test for continuous variables and the chi-square or Fisher test for categorical variables, as appropriate. The mean difference in LVEF between groups at 1 year of follow-up was also calculated and presented using 95% CI.

We constructed cluster-robustified univariable and multivariable Cox regression models to assess the association between PM implantation (pre- or post-TAVR) and the tested outcome and reported as HRs and 95% CIs while adjusting for differences between TAVR centers. All the estimates were also adjusted for age, sex, year of TAVR, and pre-TAVR risk factors: diabetes, hypertension, history of atrial fibrillation, valve type, ejection fraction, and Society of Thoracic Surgeons score. Proportional hazards were tested using Schoenfeld's residual (chi-square test) and were not significant for the primary outcome death at 1 year, 5 years, and 10 years. For the secondary outcomes, we constructed cluster-robustified (for hospital ID) Poisson models and reported the rate ratio and 95% CI. All values were then adjusted for the same covariables as previously mentioned. A sensitivity analysis of the same outcomes at 5-year follow-up is also presented. Outcome data are illustrated using the Kaplan-Meier curves.

All P values are 2-sided. To adjust for multiple comparisons, a P value <0.016 was considered statistically significant after Bonferroni correction. All statistical analyses were performed using STATA version 17.0 (StataCorp).

RESULTS

Between February 2011 and June 2022, a total of 13,360 patients (mean age 82 ± 7 years; n = 6,336 [47%] female; self-expanding valve n = 6,386 [48%]) were alive 30 days after TAVR and were included in the analysis. Of these, 1,366 (10%) patients had a pre-existing PM, while 2,028 (15%) patients required PM implantation during the first 30 days after the TAVR procedure. The patients were followed for a median of 889 days (Q1-Q3: 365-1,765 days).

BASELINE CHARACTERISTICS. An overview of the baseline characteristics of all patients and for patients with new PM implantation, with pre-existing PM implantation, and without PM implantation within 30 days after TAVR is presented in **Table 1**. In brief, patients requiring PM implantation after TAVR compared with patients without PM were older (82 ± 6 years of age vs 81 ± 7 years of age; $P < 0.001$); were less likely to be female (850 [42%] vs 4,971 [50%]; $P < 0.001$); and had higher rates of diabetes mellitus (589 [29%] vs 2,500 [25%]; $P < 0.001$), hypertension (1,672 [82%] vs 7,929 [80%]; $P = 0.003$), and atrial fibrillation (546 [34%] vs 2,424 [29%]). Patients with PM implantation had also higher Society of Thoracic Surgeons Predicted Risk of Mortality (4.7 ± 3.5 vs 4.3 ± 3.7; $P < 0.001$).

Compared with patients with pre-existing PMs, those who required PM implantation after TAVR were more likely to be female (850 [42%] vs 515 [38%]; $P = 0.02$), had lower prevalence of myocardial infarction (277 [14%] vs 207 [15%]; $P < 0.001$), and had lower prevalence of previous cardiac surgery (243 [12%] vs 292 [21%]; $P < 0.001$). Patients requiring PM after TAVR also had higher mean LVEF (56% ± 13% vs 52% ± 14%; $P < 0.001$), higher transvalvular mean gradient (41 ± 16 mm Hg vs 35 ± 16 mm Hg; $P < 0.001$), and smaller aortic valve area before TAVR (0.77 ± 0.24 cm^2 vs 0.8 ± 0.3 cm^2; $P < 0.001$) compared with patients with pre-existing PM.

PROCEDURAL DATA. **Table 2** provides an overview of the procedural characteristics. Patients who required PM implantation compared with those without PM implantation after TAVR more often received self-expanding (1,059 [52%] vs 4,617 [46%]; $P < 0.001$) and mechanically expanding (ie, Lotus [Boston Scientific]) (118 [5.8%] vs 182 [1.8%]; $P < 0.001$) valves. While femoral access was similar between groups, transapical access (59 [2.9%] vs 472 [4.7%]; $P < 0.001$) was significantly less often used in patients with new PM implantation after TAVR compared with patients with no PM.

OUTCOME DATA AT 1-YEAR FOLLOW-UP. At 1 year, patients requiring PM implantation after TAVR had higher unadjusted overall mortality (HR: 1.28; 95% CI: 1.15-1.41; $P < 0.001$) (**Central Illustration**) and cardiovascular mortality rates compared with patients with no PM (HR: 1.40; 95% CI: 1.20-1.62; $P < 0.001$) (**Central Illustration**). After multivariable adjustment for age, sex, year of TAVR and pre-TAVR risk factors, diabetes, hypertension, history of atrial fibrillation, ejection fraction, Society of Thoracic Surgeons score, and type of valve implanted, the difference in overall mortality remained significant (aHR: 1.15; 95% CI: 1.05-1.26; $P = 0.002$), as did the cardiovascular mortality (aHR: 1.25; 95% CI: 1.06-1.46; $P = 0.006$).

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TABLE 1 Baseline Characteristics

	All Patients Alive at 30 d	No PM	Pre-PM	Post-PM	*P* Value		
					No PM vs Pre-PM	No PM vs Post-AMI	Pre-PM vs Post-PM
Age, y	n = 13,360, 81.75 ± 6.55	n = 9,966, 81.48 ± 6.59	n = 1,366, 82.56 ± 6.73	n = 2,028, 82.49 ± 6.14	<0.001	<0.001	0.75
Female	n = 13,360, 6,336 (47.43)	n = 9,966, 4,971 (49.88)	n = 1,366, 515 (37.70)	n = 2,028, 850 (41.91)	<0.001	<0.001	0.02
Body mass index, kg/m^2	n = 13,349, 26.81 ± 5.07	n = 9,956, 26.76 ± 5.08	n = 1,365, 26.71 ± 4.74	n = 2,028, 27.15 ± 5.23	0.80	0.002	0.01
Diabetes mellitus	n = 13,360, 319 (2.39)	n = 9,966, 237 (2.38)	n = 1,366, 38 (2.78)	n = 2,028, 44 (2.17)	0.09	<0.001	0.26
Dialysis	n = 13,360, 3,461 (25.91)	n = 9,966, 2,500 (25.09)	n = 1,366, 372 (27.23)	n = 2,028, 589 (29.04)	0.08	0.93	0.12
Arterial hypertension	n = 13,357, 262 (1.96)	n = 9,964, 189 (1.90)	n = 1,365, 36 (2.64)	n = 2,028, 37 (1.82)	0.15	0.003	0.39
Dyslipidemia	n = 13,357, 10,711 (80.19)	n = 9,963, 7,929 (79.58)	n = 1,366, 1,110 (81.26)	n = 2,028, 1,672 (82.45)	0.31	0.08	0.70
COPD	n = 13,359, 7,467 (55.89)	n = 9,966, 5,523 (55.42)	n = 1,366, 777 (56.88)	n = 2027, 1,167 (57.57)	0.16	0.56	0.11
History of stroke	n = 13,359, 1,426 (10.67)	n = 9,965, 1,070 (10.74)	n = 1,366, 129 (9.44)	n = 2,028, 227 (11.19)	<0.001	0.32	0.03
Peripheral artery disease	n = 13,360, 1,546 (11.57)	n = 9,966, 1,106 (11.10)	n = 1,366, 199 (14.57)	n = 2,028, 241 (11.88)	0.001	0.36	0.001
Coronary artery disease	n = 13,359, 2,064 (15.45)	n = 9,965, 1,517 (15.22)	n = 1,366, 255 (18.67)	n = 2,028, 292 (14.40)	<0.001	0.30	<0.001
History of myocardial infarction	n = 13,359, 7,376 (55.21)	n = 9,966, 5,408 (54.26)	n = 1,365, 842 (61.68)	n = 2,028, 1,126 (55.52)	<0.001	0.01	0.23
History of PCI	n = 13,360, 1,646 (12.32)	n = 9,966, 1,162 (11.66)	n = 1,366, 207 (15.15)	n = 2,028, 277 (13.66)	<0.001	0.89	<0.001
History of cardiac surgery	n = 13,360, 3,751 (28.08)	n = 9,966, 2,736 (27.45)	n = 1,366, 462 (33.82)	n = 2,028, 553 (27.27)	<0.001	0.66	<0.001
Atrial fibrillation	n = 13,360, 1,768 (13.23)	n = 9,966, 1,233 (12.37)	n = 1,366, 292 (21.38)	n = 2,028, 243 (11.98)	<0.001	<0.001	<0.001
Cardiogenic shock (Killip 4)	n = 10,956, 3,502 (31.96)	n = 8,257, 2,424 (29.36)	n = 1,087, 532 (48.94)	n = 1,612, 546 (33.87)	0.84	0.73	0.10
Syncope	n = 13,346, 1,242 (9.31)	n = 9,959, 891 (8.95)	n = 1,362, 157 (11.53)	n = 2,025, 194 (9.58)	0.003	0.37	0.08
LVEF, %	n = 12,871, 56.04 ± 13.06	n = 9,600, 56.63 ± 12.79	n = 1,315, 51.95 ± 14.48	n = 1,956, 55.87 ± 12.83	<0.001	0.02	<0.001
Aortic valve area, cm^2	n = 12,489, 0.78 (0.60-0.90)	n = 9,318, 0.77 (0.60-0.90)	n = 1,270, 0.80 (0.63-0.90)	n = 1,901, 0.80 (0.60-0.90)	<0.001	0.116	0.014
Mean gradient, mm Hg	n = 12,960, 41.28 ± 16.37	n = 9,676, 42.07 ± 16.48	n = 1,316, 36.03 ± 15.59	n = 1,968, 40.91 ± 15.70	<0.001	0.004	<0.001
STS PROM	n = 13,360, 3.40 (2.20-5.35)	n = 9,966, 3.30 (2.12-5.19)	n = 1,366, 3.89 (2.59-6.00)	n = 2,028, 3.70 (2.33-5.80)	<0.001	<0.001	0.004

Values are mean ± SD, n (%), or median (Q1-Q3). *P* values are from pairwise analyses of variance (*F* test) or Fisher test. All patients alive at 30 days post-TAVR with TAVR performed between February, 15, 2011, and June 30, 2022, are included.

COPD = chronic obstructive pulmonary disease; LVEF = left ventricular ejection fraction; PCI = percutaneous coronary intervention; PM = pacemaker; Pre-PM = previous pacemaker implant; Post-PM = pacemaker implant after valve procedure; STS = Society of Thoracic Surgeons; PROM = Predicted Risk of Mortality; TAVR = transcatheter aortic valve replacement.

Compared with patients with pre-existing PMs, the outcome rates to patients requiring PM implantation after TAVR were similar (overall mortality, aHR: 0.97; 95% CI: 0.77-1.23; $P = 0.82$; cardiovascular mortality, aHR: 0.9; 95% CI: 0.63-1.29; $P = 0.57$).

A LVEF decline of ≥10% at 1-year follow-up was more common in patients requiring PM implantation after TAVR compared with patients without PM implantation (aHR: 1.84; 95% CI: 1.66-2.03; $P < 0.001$) and with pre-existing PM implantation (aHR: 1.59; 95% CI: 1.23-2.07; $P < 0.001$). These findings persisted after multivariable adjustment (aHR: 1.88; 95% CI: 1.72-2.05; $P < 0.001$; and aHR: 1.37; 95% CI: 1.11-1.69; $P = 0.003$, respectively).

Patients requiring PM implantation after TAVR were more likely to have HF in NYHA functional class III or IV at 1 year compared with patients without PM implantation (HR: 1.27; 95% CI: 1.04-1.55; $P < 0.001$) but had similar rates as those with pre-existing PM (HR: 0.96; 95% CI: 0.75-1.22; $P = 0.73$). These results were consistent after multivariable adjustment (aHR: 1.27; 95% CI: 1.05-1.54; $P = 0.015$; and aHR: 0.98; 95% CI: 0.76-1.26; $P = 0.86$).

A composite outcome of all-cause death, LVEF decline ≥10%, and NYHA functional class III or IV at 1 year occurred more often in patients requiring PM implantation after TAVR compared with patients without PM implantation (HR: 1.37; 95% CI: 1.3-1.45; $P < 0.001$). Multivariable adjustment confirmed these results (aHR: 1.34; 95% CI: 1.28-1.41; $P < 0.001$).

Compared with patients with pre-existing PM, those requiring PM implantation after TAVR had similar rates of all-cause death, LVEF decline ≥10%, and NYHA functional class III or IV at 1 year (aHR: 1.12; 95% CI: 0.01-1.24; $P = 0.03$) but had higher rates of the composite outcome of cardiovascular death, LVEF decline ≥10%, and NYHA functional class III or IV at 1 year (HR: 1.27,; 95% CI: 1.04-1.55; $P = 0.016$),

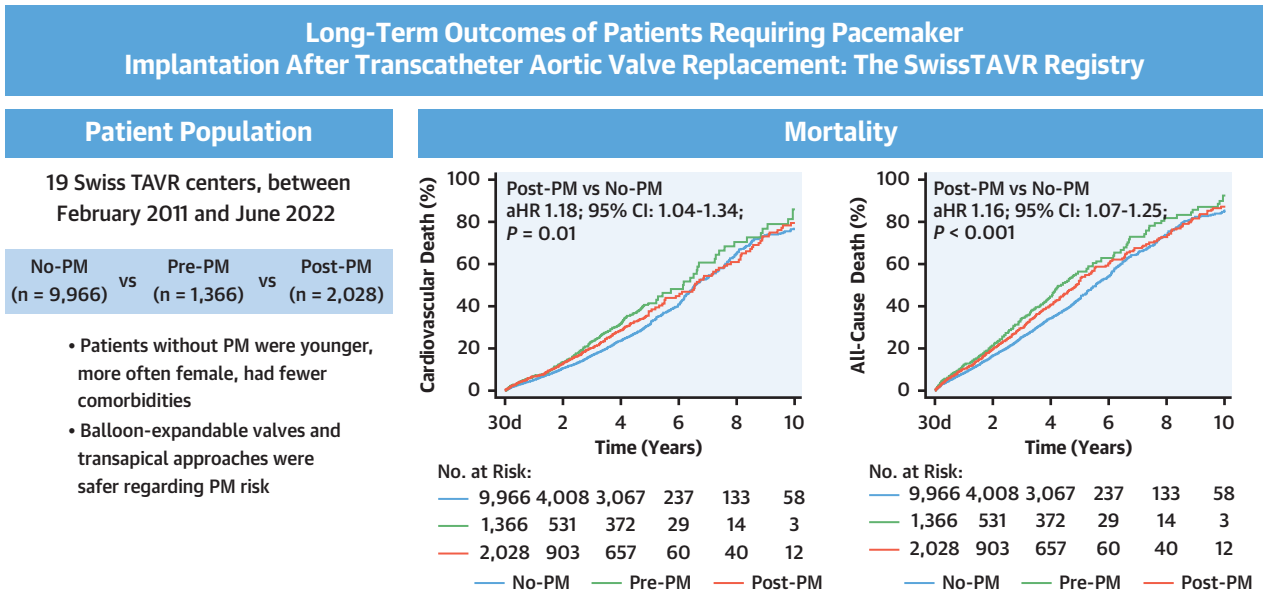
TABLE 2 Procedural Characteristics

	All Patients Alive at 30 d (N = 13,360)	No PM (n = 9,966)	Pre-PM (n = 1,366)	Post-PM (n = 2,028)	*P* Value		
					No PM vs Pre-PM	No PM vs Post-PM	Pre-PM vs Post-PM
Access (main)	n = 13,359	n = 9,966	n = 1,365	n = 2,028	0.09	<0.001	0.03
Femoral	12,447 (93)	9,277 (93)	1,258 (92)	1,912 (94)	0.21	0.05	0.02
Transapical	595 (4.5)	472 (4.7)	64 (4.7)	59 (2.9)	1.000	<0.001	0.01
Subclavian	130 (1.0)	81 (0.8)	19 (1.4)	30 (1.5)	0.04	0.01	0.88
Direct aortic	96 (0.7)	71 (0.7)	9 (0.7)	16 (0.8)	1.000	0.67	0.84
Other	91 (0.7)	65 (0.7)	15 (1.1)	11 (0.5)	0.08	0.65	0.07
Balloon valvuloplasty (no)	n = 13,359, 5,981 (44.8)	n = 9,966, 4,367 (43.8)	n = 1,366, 697 (51.0)	n = 2,027, 917 (45.2)	<0.001	0.25	0.001
Heart valve type	n = 13,341	n = 9,950	n = 1,365	n = 2,026			
Balloon expanding	6,617 (49.6)	5,151 (51.8)	617 (45.2)	849 (41.9)	<0.001	<0.001	0.06
Self-expanding	6,386 (47.9)	4,617 (46.4)	710 (52.0)	1059 (52.3)	<0.001	<0.001	0.89
Mechanicallly expanding (Lotus valve)	338 (2.5)	182 (1.8)	38 (2.8)	118 (5.8)	0.02	<0.001	<0.001
Moderate or severe aortic regurgitation at discharge	n = 13,322, 459 (3.4)	n = 9,934, 318 (3.2)	n = 1,365, 52 (3.8)	n = 2,023, 89 (4.4)	0.26	0.01	0.43
Postprocedure mean gradient, mm Hg	n = 13,039, 8.28 ± 5.16	n = 9,709, 8.43 ± 5.22	n = 1,341, 7.97 ± 5.46	n = 1,989, 7.74 ± 4.56	0.003	<0.001	0.18

Values are n (%) or mean ± SD. *P* values are from pairwise analyses of variance (*F* test) or Fisher test.

APT = antiplatelet agent; DAPT = dual antiplatelet therapy; OAC = oral anticoagulation; SAPT = single antiplatelet therapy; other abbreviations as in Table 1.

CENTRAL ILLUSTRATION 10-Year Outcomes Following Pacemaker Placement After TAVR



- In this multicentric, prospective, observational study, patients implanted with a PM within 30 days after TAVR had worse all-cause mortality and cardiovascular mortality than patients without PM implant at 1, 5, and 10 years.
- Patients implanted with a PM within 30 days after TAVR had higher rates of LVEF worsening ≥10% and more HF symptoms at 1 year than patients without PM.

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aHR = adjusted HR; HF = heart failure; LVEF = left ventricular ejection fraction; PM = pacemaker; SwissTAVI = Swiss Transcatheter Aortic Valve Implantation; TAVR = transcatheter aortic valve replacement.

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TABLE 3 Unadjusted and Adjusted Secondary Outcomes at 1 Year

	Unadjusted Secondary Outcomes at 1 y					
	Pre-PM vs No PM		Post-PM vs No PM		Post-PM vs Pre-PM	
	Unadjusted Rate Ratio (95% CI)	*P* Value	Unadjusted Rate Ratio (95% CI)	*P* Value	Unadjusted Rate Ratio (95% CI)	*P* Value
Death, LVEF decline ≥10%, NYHA functional class III or IV	1.30 (1.20 to 1.42)	<0.001	1.37 (1.30 to 1.45)	<0.001	1.05 (0.96 to 1.16)	0.265
Cardiovascular death, LVEF decline ≥10%, NYHA functional class III or IV	1.23 (1.09 to 1.38)	<0.001	1.46 (1.36 to 1.56)	<0.001	1.19 (1.05 to 1.34)	0.004
NYHA functional class III or IV	1.33 (1.04 to 1.69)	0.021	1.27 (1.04 to 1.55)	0.017	0.96 (0.75 to 1.22)	0.736
LVEF decline ≥10%	1.15 (0.87 to 1.53)	0.330	1.84 (1.66 to 2.03)	0.000	1.59 (1.23 to 2.07)	0.000
	Mean Difference (95% CI)	*P* Value	Mean Difference (95% CI)	*P* Value	Mean Difference (95% CI)	*P* Value
LVEF % at 1 y/sample size	−5.32 (−5.98 to −4.65)	<0.001	−3.24 (−3.85 to −2.63)	<0.001	2.08 (1.08 to 3.07)	<0.001
	Adjusted Secondary Outcomes at 1 y					
	Pre-PM vs No PM		Post-PM vs No PM		Post-PM vs Pre-PM	
	Adjusted Rate Ratio (95% CI)	*P* Value	Adjusted Rate Ratio (95% CI)	*P* Value	Adjusted Rate Ratio (95% CI)	*P* Value
Death, LVEF decline ≥10%, NYHA functional class III or IV	1.15 (1.02 to 1.31)	0.023	1.34 (1.28 to 1.41)	<0.001	1.12 (1.01 to 1.24)	0.033
Cardiovascular death, LVEF decline ≥10%, NYHA functional class III or IV	1.15 (0.99 to 1.33)	0.066	1.44 (1.35 to 1.54)	<0.001	1.20 (1.05 to 1.38)	0.010
NYHA functional class III or IV	1.19 (0.90 to 1.57)	0.219	1.27 (1.05 to 1.54)	0.015	0.98 (0.76 to 1.26)	0.864
LVEF decline ≥10%	1.46 (1.13 to 1.88)	0.004	1.88 (1.72 to 2.05)	<0.001	1.37 (1.11 to 1.69)	0.003
	Mean Difference (95% CI)	*P* Value	Mean Difference (95% CI)	*P* Value	Mean Difference (95% CI)	*P* Value
LVEF % at 1 y/sample size	−2.90 (−3.40 to −2.41)	<0.001	−2.88 (−3.31 to −2.44)	<0.001	−0.20 (−0.69 to 0.30)	0.439

Patients alive without LVEF baseline or without LVEF at 1 year or without NYHA functional class assessed at 1 year were not used in the secondary outcome analyses. Proportions/denominator (% of assessed or deceased patients) with pairwise rate ratios with 95% CI were used, except for LVEF, which are mean differences is presented (*F* test). Adjusted estimates after adjusting for age, sex, year of TAVR, TAVR type of device (self-expandable, balloon expandable, or mechanically expandable) and pre-TAVR risk factors: LVEF (imputed with median value of 60% if missing), STS PROM score, diabetes, hypertension, and history of atrial fibrillation. Because atrial fibrillation was only recorded from 2016 onward, sample sizes are reduced, see number in sample/total sample size. Estimation was cluster-robustified for the hospital ID.

Abbreviations as in Table 1.

consistent after multivariable adjustment (aHR: 1.2; 95% CI: 1.05-1.38; $P = 0.01$) (Table 3).

OUTCOME DATA AT 5- AND 10-YEAR FOLLOW-UP. At 5-year follow-up, patients requiring PM implantation after TAVR had higher adjusted overall mortality rates (aHR: 1.16; 95% CI: 1.07-1.26; $P < 0.001$) and adjusted cardiovascular mortality rates (aHR: 1.18; 95% CI: 1.04-1.36; $P = 0.013$) compared with patients without PM (Table 4). These findings were consistent at 10-year follow-up, with adjusted overall mortality rates (aHR: 1.16; 95% CI: 1.07-1.25; $P < 0.001$) and adjusted cardiovascular mortality rates (aHR: 1.18; 95% CI: 1.04-1.34; $P = 0.01$) compared with patients without PM. Compared with patients with pre-existing PMs, the outcome rates for those requiring PM implantation after TAVR were similar.

DISCUSSION

In this analysis of a large, prospective, nationwide cohort study investigating the outcomes of patients undergoing permanent PM implantation after TAVR, the major findings are the following.

First, patients requiring PM implantation within 30 days after TAVR had significantly higher rates of overall and cardiovascular mortality rates compared with patients without PM implantation at 1-, 5-, and 10-year follow-up. These findings were consistent after adjusting for baseline and procedural characteristics, except for overall mortality at 1 year, which showed no difference. Second, patients who underwent PM implantation after TAVR had significantly higher adjusted rates of a LVEF decline (≥10%), HF in NYHA functional class III or IV, and a composite outcome of cardiovascular death, LVEF decline ≥10%, and HF in NYHA functional class III or IV at 1 year. These results highlight the need for novel technologies and techniques to further reduce PM rates after TAVR, implement better risk stratification strategies to target PM therapy to those patients most likely to benefit, and explore more physiological pacing methods, such as conduction system pacing, within the TAVR population. Third, compared with patients without PM implantation, patients with pre-existing PM implantation had similar outcomes regarding all-cause mortality and cardiovascular mortality during long-term follow-up.

TABLE 4 Unadjusted and Adjusted Secondary Outcomes at 5 and 10 Years

	Pre-PM vs No PM		Post-PM vs No PM		Post-PM vs Pre-PM		Pre-PM vs No PM		Post-PM vs No PM		Post-PM vs Pre-PM	
	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value	Adjusted HR (95% CI)	Adjusted *P* Value	Adjusted HR (95% CI)	Adjusted *P* Value	Adjusted HR (95% CI)	Adjusted *P* Value
At 5 y												
All-cause mortality	1.44 (1.35-1.54)	<0.001	1.25 (1.15-1.36)	<0.001	0.87 (0.77-0.97)	0.013	1.11 (0.93-1.31)	0.243	1.16 (1.07-1.26)	<0.001	1.01 (0.82-1.25)	0.907
Cardiovascular mortality	1.41 (1.26-1.57)	<0.001	1.24 (1.10-1.39)	<0.001	0.88 (0.76-1.02)	0.080	1.07 (0.88-1.31)	0.508	1.18 (1.04-1.36)	0.013	1.06 (0.82-1.36)	0.671
At 10 y												
All-cause mortality	1.42 (1.32-1.51)	<0.001	1.22 (1.13-1.33)	<0.001	0.87 (0.78-0.96)	0.007	1.10 (0.94-1.29)	0.220	1.16 (1.07-1.25)	<0.001	1.01 (0.83-1.23)	0.924
Cardiovascular mortality	1.37 (1.26-1.50)	<0.001	1.20 (1.09-1.32)	<0.001	0.88 (0.78-1.00)	0.046	1.08 (0.89-1.31)	0.454	1.18 (1.04-1.34)	0.010	1.05 (0.82-1.34)	0.716

Depicted are number of events (% from Kaplan-Meier estimates), only counting the first event per type per patient from day 31 to 365 post-TAVR. HRs from Cox regressions (with 95% CIs, *P* values from Wald tests) and adjusted HRs (with 95% CIs, *P* values from Wald tests) after adjusting for age, sex, year of TAVR, TAVR type of device (self-expandable, balloon expandable, or mechanically expandable) and pre-TAVR risk factors: LVEF (imputed with median value of 60% if missing), STS PROM score, diabetes, hypertension, and history of atrial fibrillation. Because atrial fibrillation was only recorded from 2016 onward, sample sizes are reduced, see number in sample/total sample size. Estimation was cluster-robustified for the hospital ID. Adjudicated events using VARC-2 (VARC-1 before 2013). For cohorts with TAVR procedures up to June 30, 2013: 10-year follow-up was used. For cohorts with with TAVR procedures up to June 30, 2018: 5-year follow-up was used. For cohorts with with TAVR procedures up to June 30, 2022: 1-year follow-up used. These cohorts were completed with events adjudicated and were administratively censored at 3,650, 1,825, and 365 days, respectively.

VARC = Valve Academic Research Consortium; other abbreviations as in Table 1.

Our findings shed new light on the conflicting evidence from prior studies regarding the impact of PM implantation after TAVR on clinical outcomes. A recent meta-analysis encompassing 42,927 patients (with 20,872 patients enrolled in 1 single study) showed an increased risk of all-cause death and HF hospitalization at 1 year in patients requiring a PM after TAVR but did not show a detrimental impact on cardiovascular death.[22] In contrast, the SWEDEHEART (the Swedish Web-system for Enhancement and Development of Evidence-based care in Heart disease Evaluated According to Recommended Therapies) observational study including 3,420 patients over a median follow-up of 2.7 years found no difference in long-term survival or HF between patients who did and did not undergo PM implantation after TAVR, though this study lacked information on LVEF or NYHA functional class during follow-up.[9] Auffret et al[8] used data from the French national health single-payer claims database and reported among 36,549 patients undergoing TAVR an increased risk of all-cause death and HF hospitalization among those requiring PM implantation during 5-year follow-up. The weakness of this study included a potential coding-related bias and no data regarding causes of death.[8] The variation in results across studies may be explained by the heterogeneity of patients analyzed, the indications for PM therapy (prophylactic vs emergent), and follow-up durations.

The scientific evidence strongly supports the notion that the deleterious impact of right ventricular pacing contributes to increased HF risk. The multicentric, prospective PACE TAVR (PACE Transcatheter aortic valve implantation) study[10] identified high right ventricular pacing burden as a major contributing factor to cardiovascular mortality and HF events. Similarly, Hochstadt et al[23] reported a significant decline in LVEF at 1 year in 1,489 patients with PM implantation after TAVR. The present study supports this hypothesis, demonstrating similar prognosis between patients with pre-existing PM therapy and those with new PM implantation after TAVR during 5 and 10 years of follow-up, but with higher rates of LVEF decline in patients with PM therapy after TAVR at 1 year of follow-up.

Patients with pre-existing PM implantation differ from those requiring PM implantation after TAVR in several key ways. First, patients with pre-existing PM implantation may have received pacing due to sick sinus syndrome, often with preserved atrioventricular conduction, which helps avoid the detrimental effects of right ventricular pacing. Second, these patients likely experienced a more gradual progression of conduction system disease, while those requiring PM implantation after TAVR typically face acute and severe conduction disturbances caused by the procedure itself.

These findings are clinically important as TAVR continues to expand to younger, lower-risk patients with longer life expectancy. In these patients, PM implantation could have substantial long-term deleterious effects.[8-14] Novel PM implantation techniques, such as conduction system pacing, are gaining increased popularity and may help preventing PM-induced cardiomyopathy and thereby potentially improving outcomes.[24] However, the potential benefits of conduction system pacing in the TAVR population remain to be demonstrated. Future

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studies are needed to better characterize patients who are likely to benefit from more physiological pacing.

The current study has several strengths. First, this is a large, prospective, multicenter cohort study of all consecutive patients undergoing TAVR in Switzerland with prespecified long-term follow-up, allowing for more accurate outcome predictions. Second, the availability of detailed baseline and procedural data allowed for adjusting for many known potential confounders. Third, by including only patients who received PM therapy within 30 days of the TAVR procedure, we minimized the potential for immortal time bias.

STUDY LIMITATIONS. First, information regarding pacing dependency and ventricular pacing burden during follow-up was not available. Second, despite adjusting for numerous clinically relevant variables, observational studies remain subject to residual confounding. Third, we did not assess the impact of pacemaker types and indications on clinical outcomes. Fourth, while TAVR indications are expanding to younger and low-risk patients, the cohort investigated in this study primarily consists of higher-risk patients, which may limit the generalizability of our findings to these newer, low-risk groups. This gap in the literature is crucial to address as TAVR becomes more common in this patient subgroup.

CONCLUSIONS

In this large nationwide registry of TAVR recipients, patients receiving PM implantation within 30 days after TAVR had significantly higher adjusted rates of cardiovascular mortality at 1 year, as well as higher rates of both all-cause and cardiovascular mortality at 5- and 10-year follow-up. These findings underscore the need for further research aimed at reducing the incidence of PM implantation after TAVR and mitigating its associated risks, especially in younger and to lower-risk patients.

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PERSPECTIVES

WHAT IS KNOWN? The most common post-TAVR complication remains conduction disturbances, potentially leading to PM implantation.

WHAT IS NEW? Patients requiring cardiac pacing after TAVR experience significantly increased overall and cardiovascular mortality rates at 1, 5, and 10 years of follow-up compared with patients without PM implantation. Patients with PM implantation after TAVR are more likely to experience a decline in LVEF and have a higher incidence of heart failure during long-term follow-up.

WHAT IS NEXT: Novel PM implantation techniques, such as conduction system pacing, may help avoiding PM-induced cardiomyopathy. However, this has yet to be demonstrated in the TAVR population.

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